

Numerical Simulations of PBH Formation: Delayed FOPTs and 3+1D Radiation-Era Collapse

Zhuan Ning (宁专), UCAS

Based on: *JCAP 07 (2026) 073*; *arXiv: 2608.13206*

Co-authors: Xiang-Xi Zeng, Rong-Gen Cai, Shao-Jiang Wang, Chul-Moon Yoo

Dynamics of Primordial Black Hole Formation III

September 8, 2026 @ Rikkyo University, Tokyo



Outline

▣ PBH formation from false vacuum domains

- ✓ Numerical simulation setups
- ✓ Dynamics of false vacuum domains
- ✓ Thresholds for type A/B PBHs

▣ PBH formation from curvature perturbations in 3D

- ✓ Numerical simulation setups
- ✓ Cosmological gauge strategy
- ✓ Numerical validation and physics results

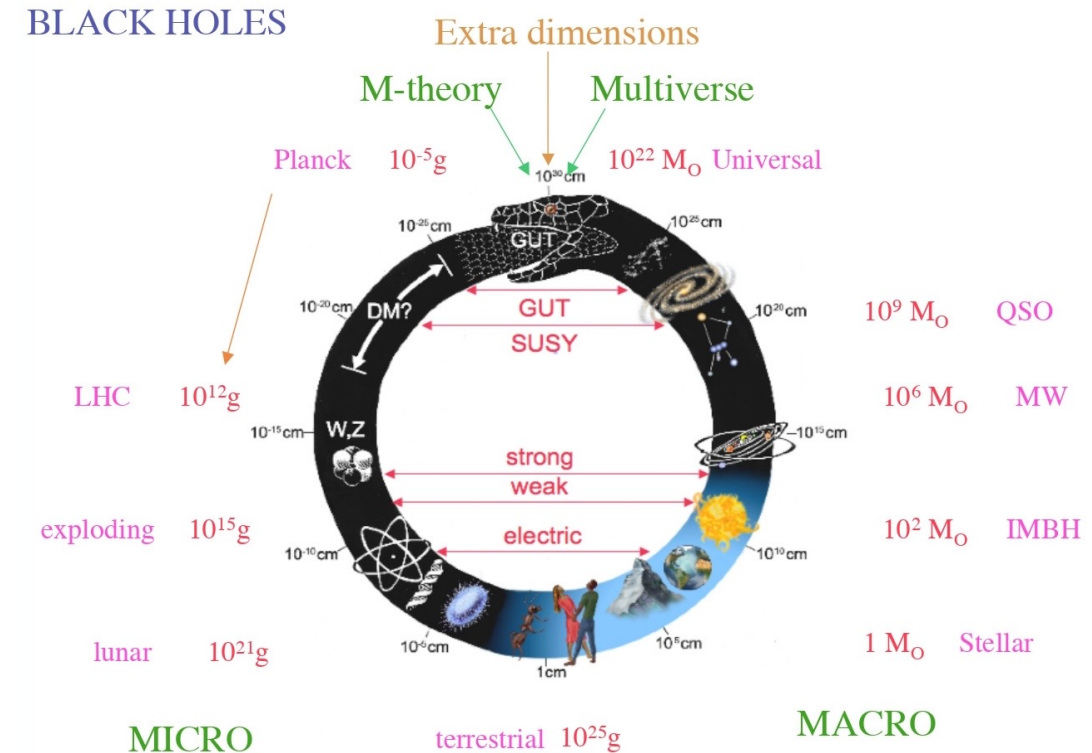
Primordial black holes (PBHs)

➤ Why PBHs?

- Probe of small-scale primordial perturbations
- Candidate for Cold Dark Matter
- Progenitors of LIGO-Virgo-KAGRA binary mergers
- Seeds for supermassive black holes

➤ Formation mechanisms:

- **Collapse of curvature perturbations**
- (P)reheating dynamics
- Collapse of topological defects (cosmic strings, domain walls)
- **First-order phase transitions (FOPTs)**
-



[Credit: Bernard Carr]

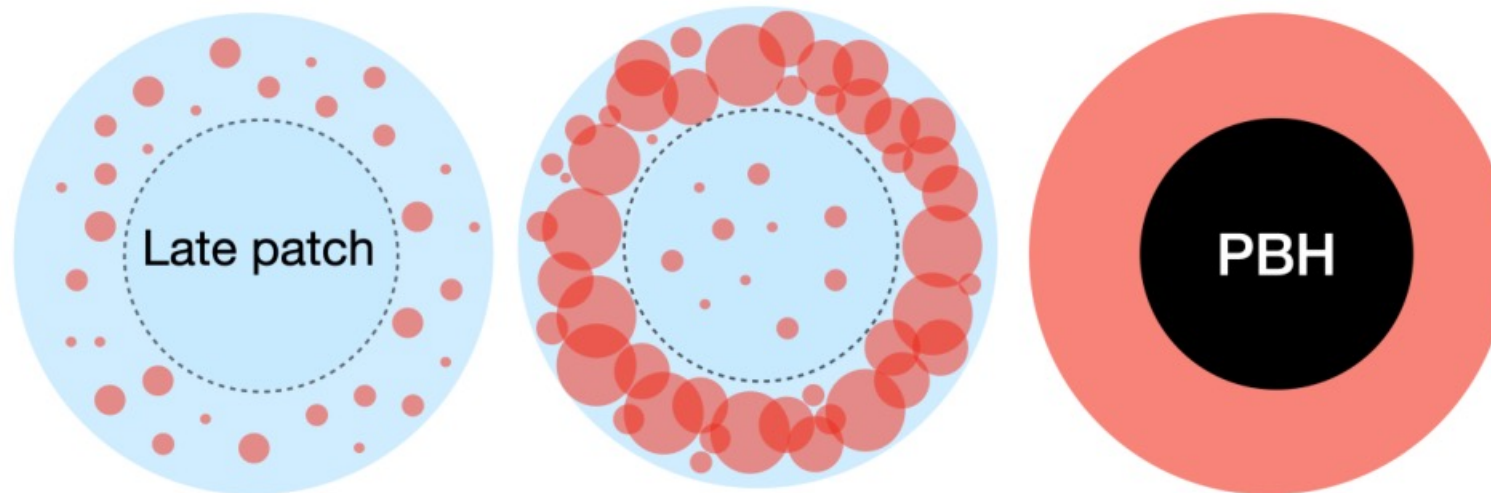
PBH Formation from False Vacuum Domains

PBHs from delayed FOPTs

[Kanemura and Wang's talk]

- Bubble nucleation is stochastic
- Strongly supercooled phase transitions tend to be **delayed**
- Some regions remain in the **false vacuum domains (FVDs)**
 - Outside FVD: Radiation dominated (dilutes as $\rho_R \sim a^{-4}$)
 - Inside FVD: Vacuum energy dominated (**constant** ρ_V)

FVDs can also arise from
inflation [Garriga and Joana's talk]
and inverted bubbles [Murai's talk]



- Result: A **large overdensity** develops inside the FVD, potentially collapsing into a PBH

PBH formation conditions



➤ **Key question:** Under what conditions does an FVD collapse into a PBH?

➤ Traditional **critical overdensity criterion**:

- $\delta = \frac{\rho_{in} - \rho_{out}}{\rho_{out}} > \delta_c \sim 0.45$
- Adopted from curvature perturbation collapse during radiation domination

➤ Recent criticisms:

- [Flores, Kusenko, Sasaki PRD2024]: Only expanding solutions exist; need **Schwarzschild criterion** $2M/R > 1$
- [Franciolini et al. 2503.01962; Wang et al. 2601.14412]: Gauge dependence issue

Dynamics of FVDs

[Kanemura's talk]

[Hashino, Kanemura, Takahashi,
Tanaka, Yoo JCAP2025]

➤ Previous work:

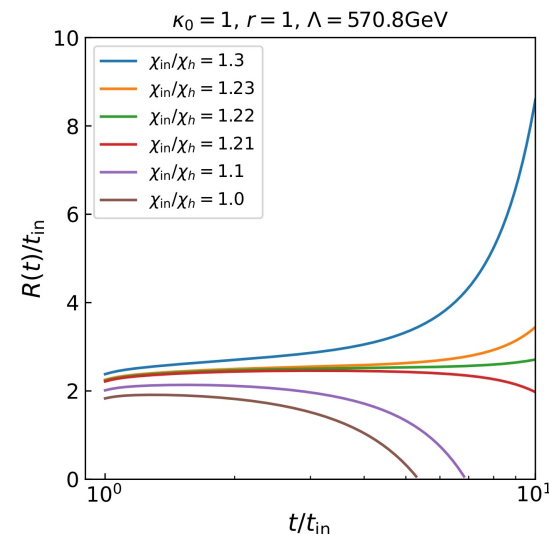
- FVD is spherically symmetric; **Thin-wall**; No new nucleation
- Israel's junction conditions to match interior and exterior metrics

$$\ddot{\chi} + (4 - 3a^2\dot{\chi}^2)H\dot{\chi} + \frac{2}{a^2\chi}(1 - a^2\dot{\chi}^2) = \left(\frac{3\sigma}{4M_{\text{pl}}^2} - \frac{\rho_V}{\sigma} \right) \frac{(1 - a^2\dot{\chi}^2)^{3/2}}{a}$$

➤ Two kinds of solutions:

- FVD boundary radius tends to infinity --- supercritical PBH
- FVD boundary radius reaches zero --- **cannot distinguish subcritical PBH or dispersal!**

Our goal: Perform **fully nonlinear numerical simulations** of coupled gravity-scalar-fluid systems to study the dynamics and determine robust formation criteria



The model

- Coupled gravity-scalar-fluid system

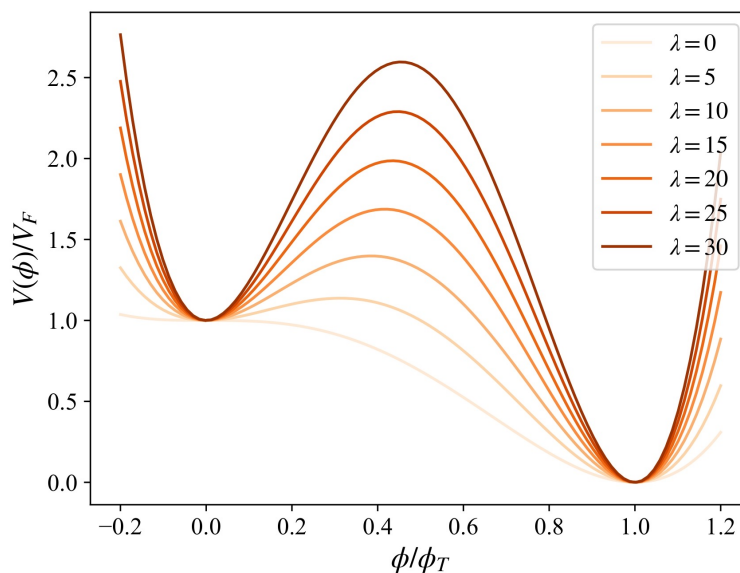
$$S = \int d^4x \sqrt{-g} \left[\frac{{}^{(4)}R}{16\pi} - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) + \mathcal{L}_{\text{fluid}} \right]$$

- Spherically symmetric metric (geodesic slicing $A = 1$)

$$ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega^2$$

- Scalar field: phenomenological potential $V(\phi) = \left[1 + \lambda \left(\frac{\phi}{\phi_T} \right)^2 - (2\lambda + 4) \left(\frac{\phi}{\phi_T} \right)^3 + (\lambda + 3) \left(\frac{\phi}{\phi_T} \right)^4 \right] (V_F - V_T) + V_T$

- True vacuum: $\phi = \phi_T, V_T = 0$
- False vacuum: $\phi = 0, \rho_V = V_F$
- $\lambda \rightarrow \infty$: Thin-wall limit
- $\lambda \rightarrow 0$: Thick-wall limit



- Fluid: radiation fluid ($p = \rho/3$), interacting with scalar only via gravity

Initial conditions & Numerical implementation

- Superhorizon FVD embedded in radiation FLRW background

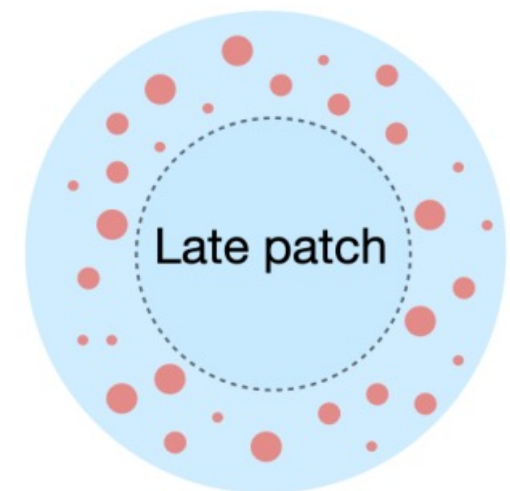
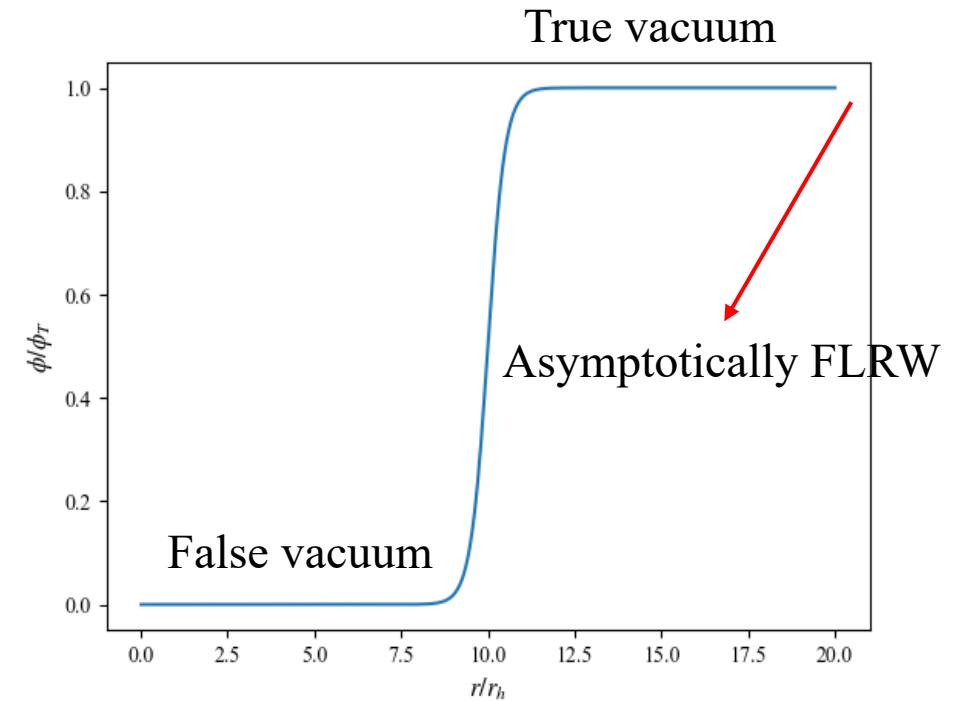
$$\phi(t_i, r) = \frac{\phi_T}{2} \left[1 + \tanh \left(\frac{r - r_i}{l} \right) \right]$$

- Initial wall thickness $l = \frac{r_i}{10}$
- To satisfy Hamiltonian constraint

$$\rho(t_i, r) = \rho_b(t_i) - \rho_\phi(t_i, r)$$

- Numerical Implementation

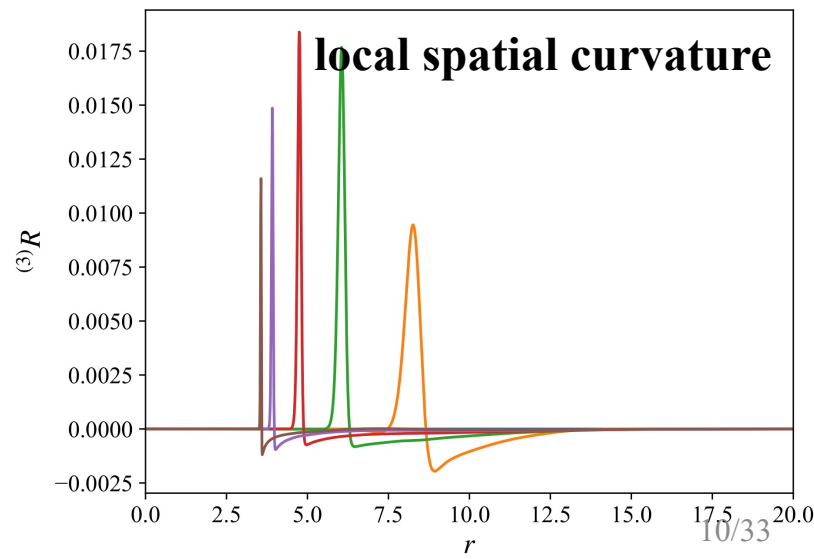
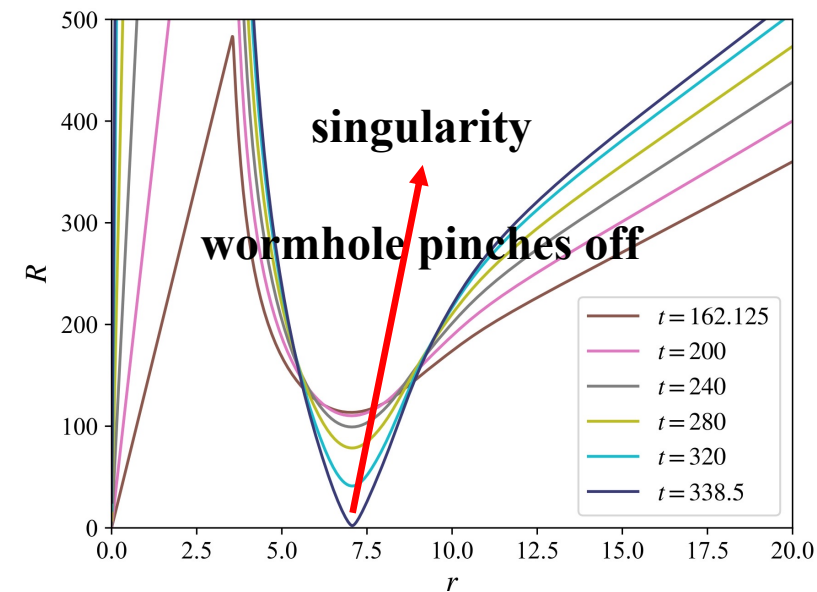
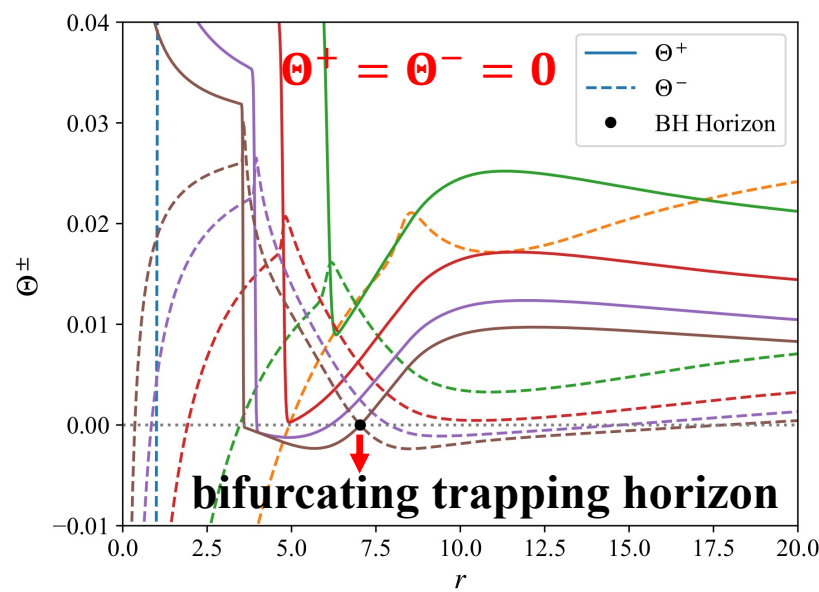
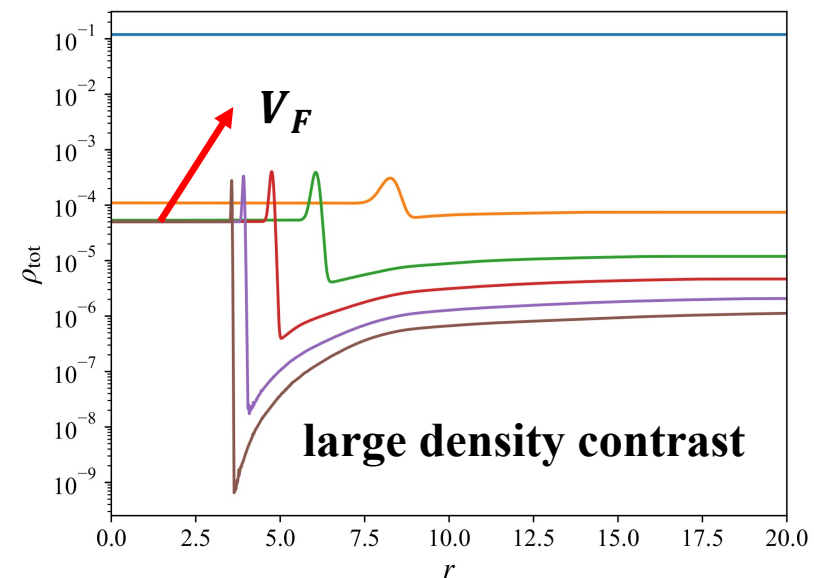
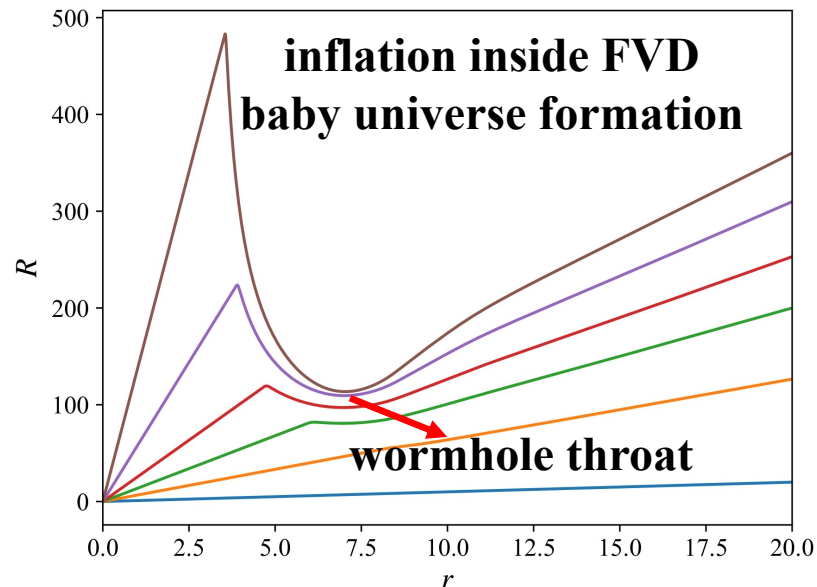
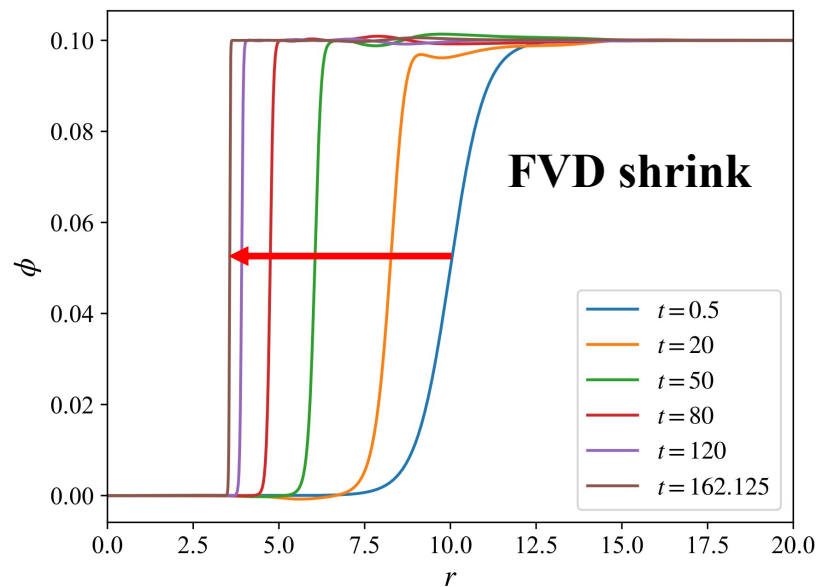
- Finite-difference spatial discretization
- Runge-Kutta time integration
- Adaptive mesh refinement to resolve bubble wall



Outcome 1: Type B PBH (Supercritical)

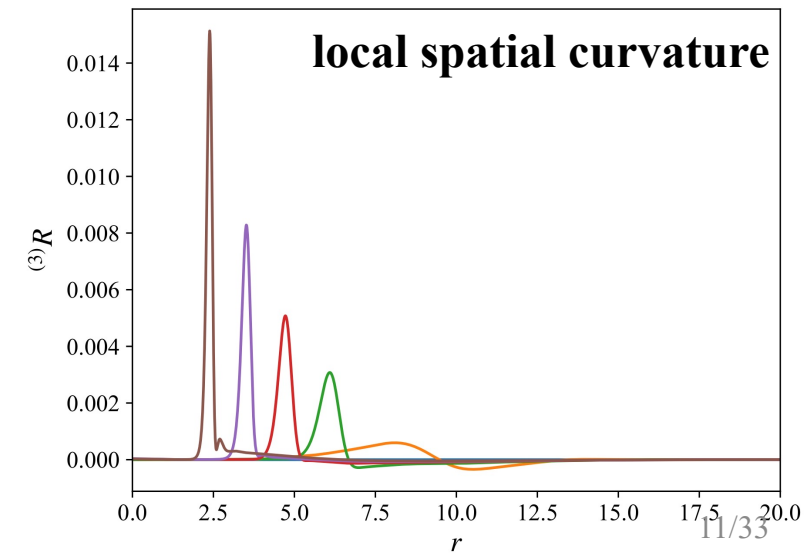
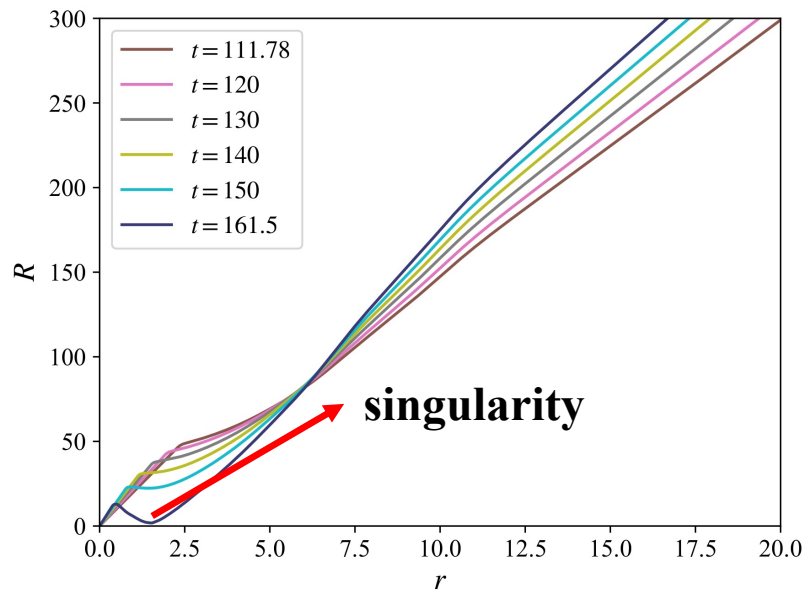
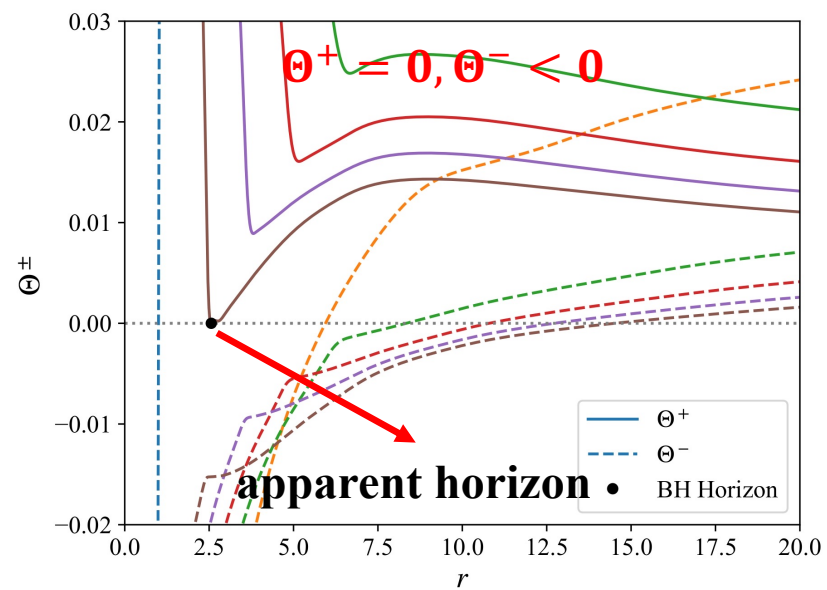
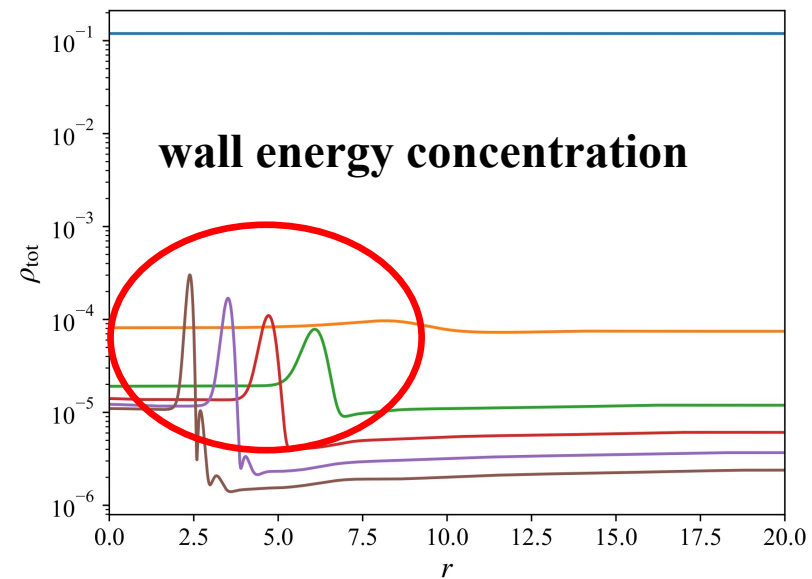
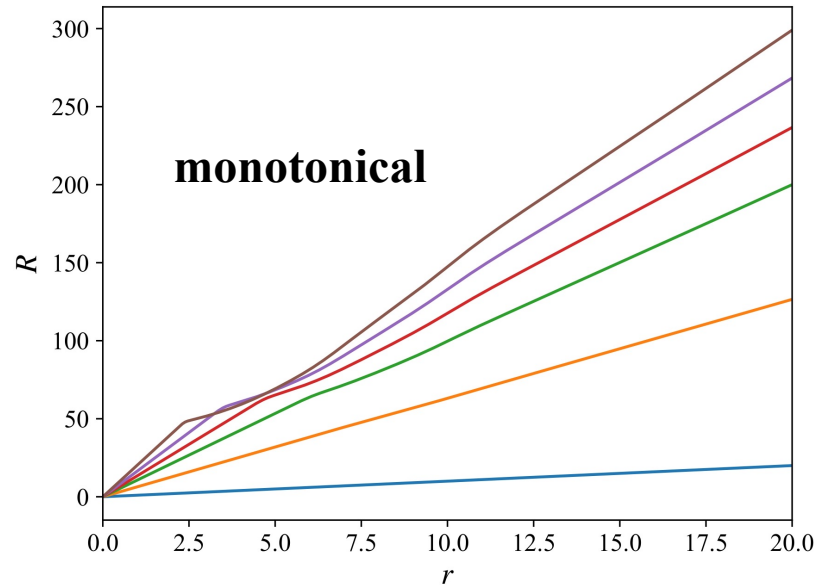
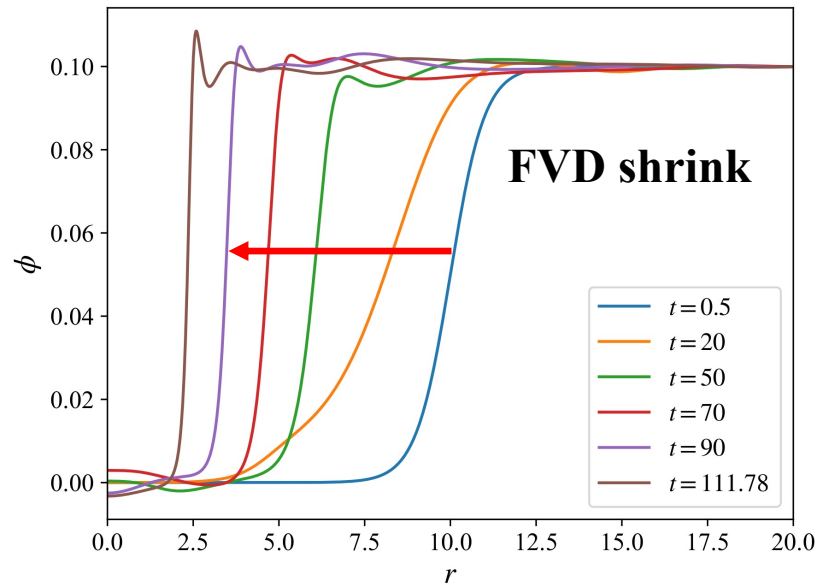
$$\phi_T = 0.1, \quad \lambda = 20, \quad r_i = 10$$

$$V_F = 5 \times 10^{-5} H_i^2$$



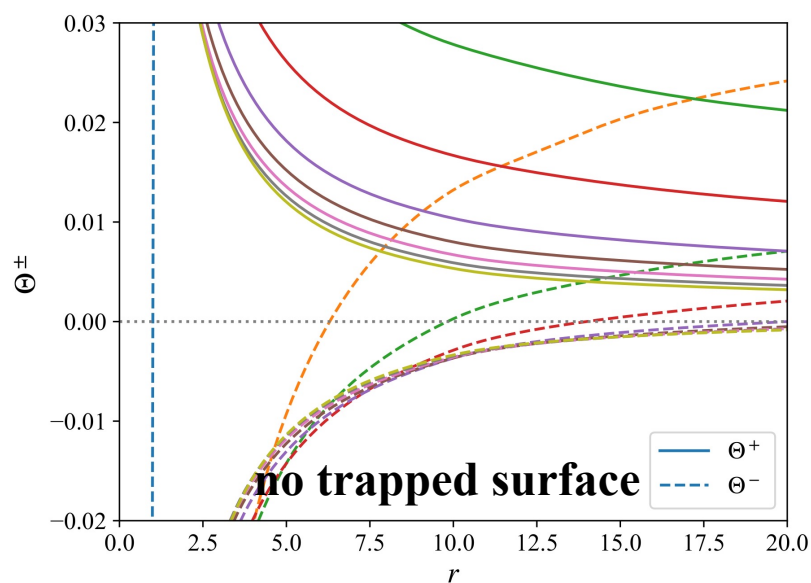
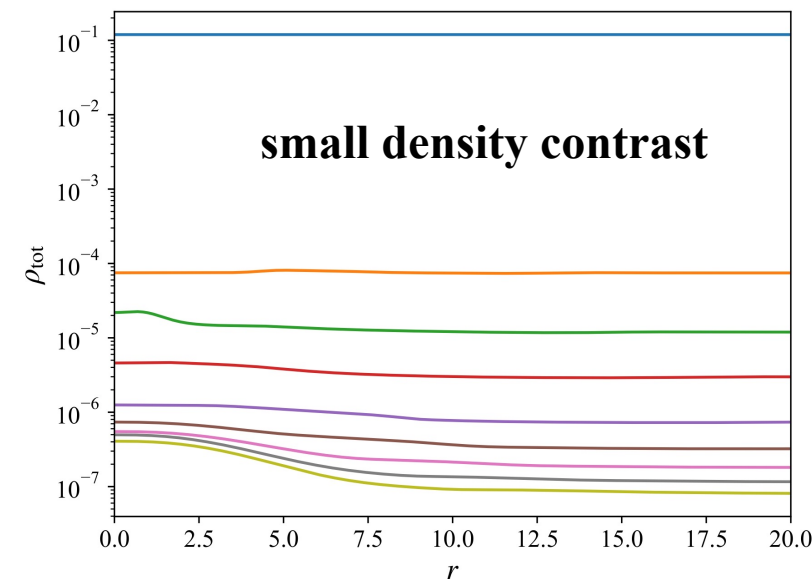
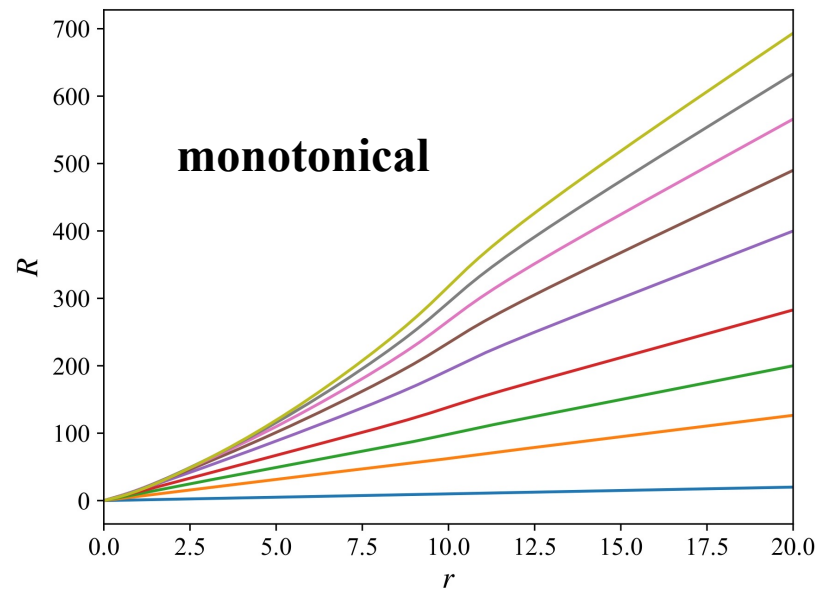
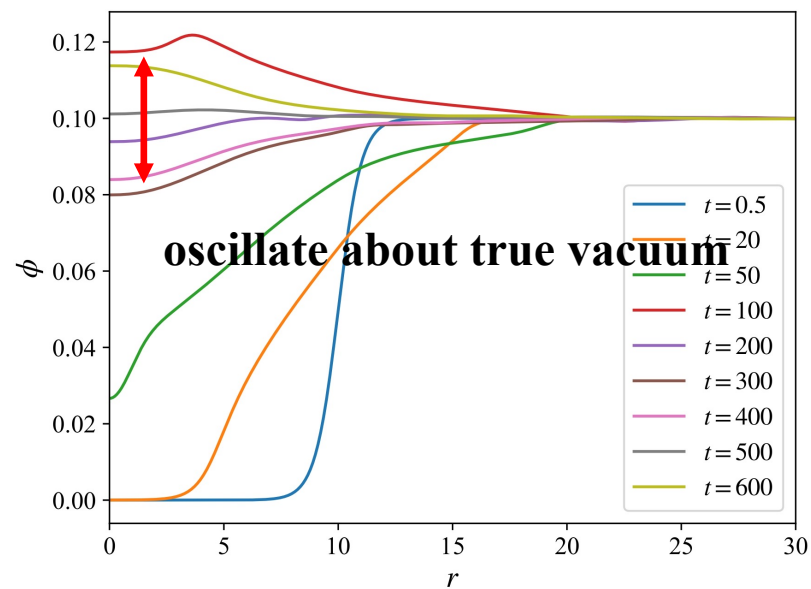
Outcome 2: Type A PBH (Subcritical)

$$V_F = 1 \times 10^{-5} H_i^2$$

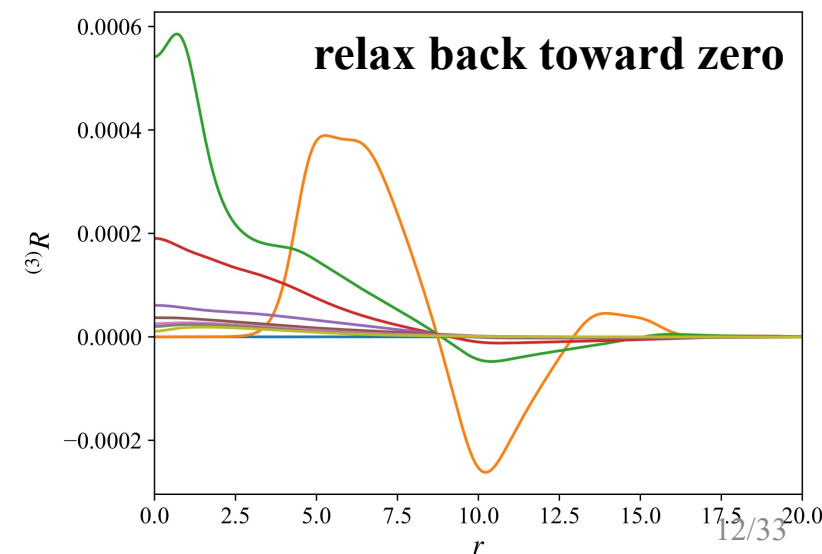


Outcome 3: No PBH Formation (Dispersal)

$$V_F = 5 \times 10^{-7} H_i^2$$



- Type B: Supercritical PBH with an interior baby universe
- Type A: Subcritical PBH from direct wall collapse
- Dispersal: No PBH formed (phase transition completes and scalar wave radiation)



Defining the criteria

- Horizon-based diagnostics are equivalent to the Schwarzschild condition in spherical symmetry

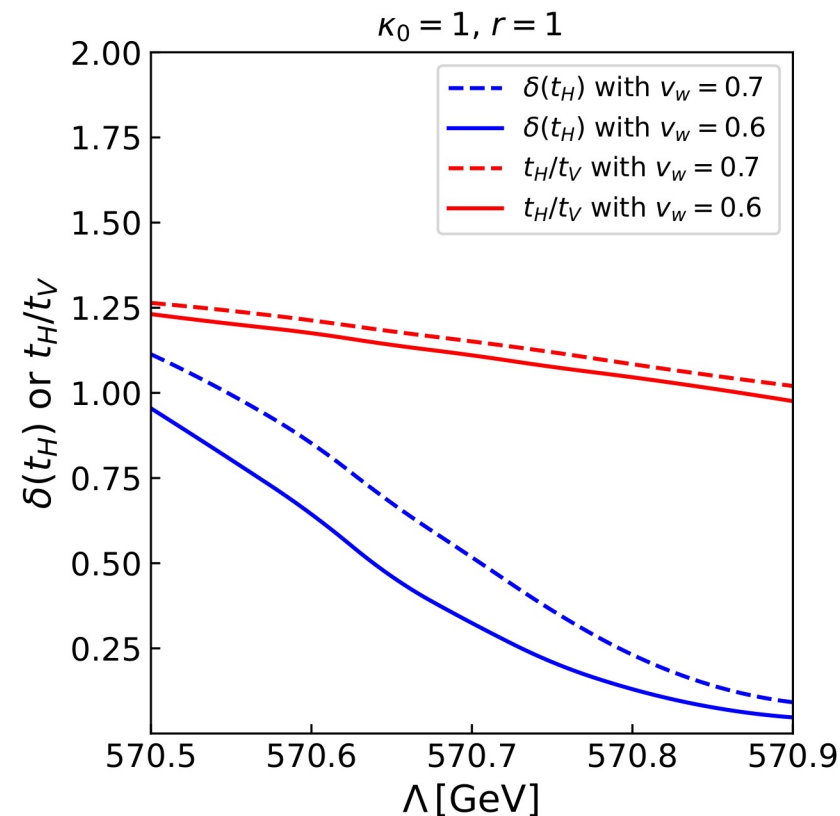
$$\Theta^+ \Theta^- = \frac{2}{R^2} (U^2 - \Gamma^2) = 0 \implies \frac{2M}{R} = 1$$

1. Time scale ratio t_H/t_V

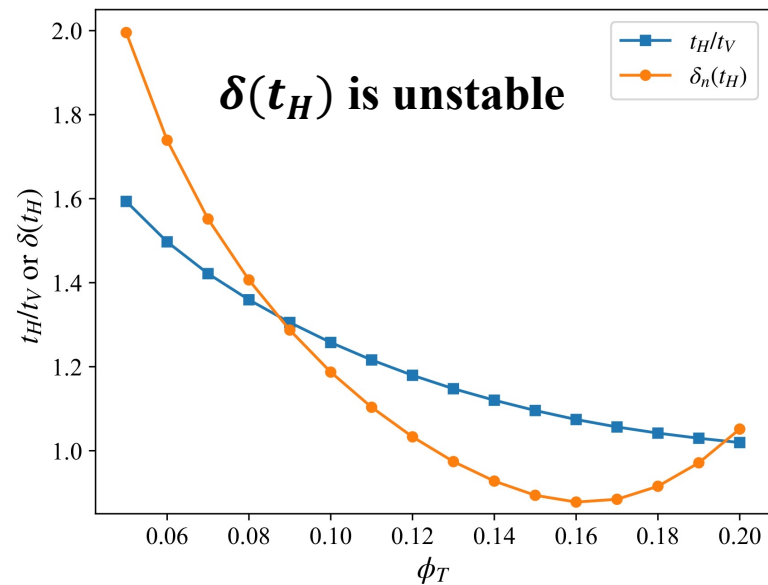
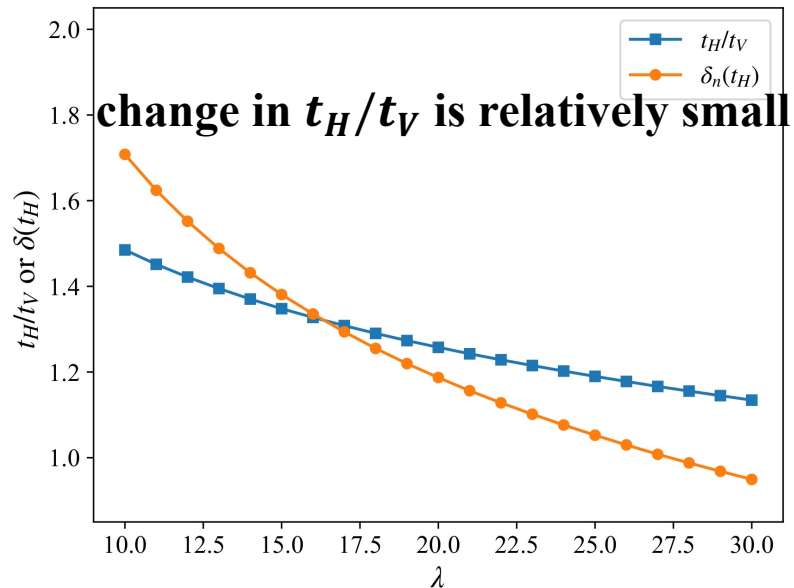
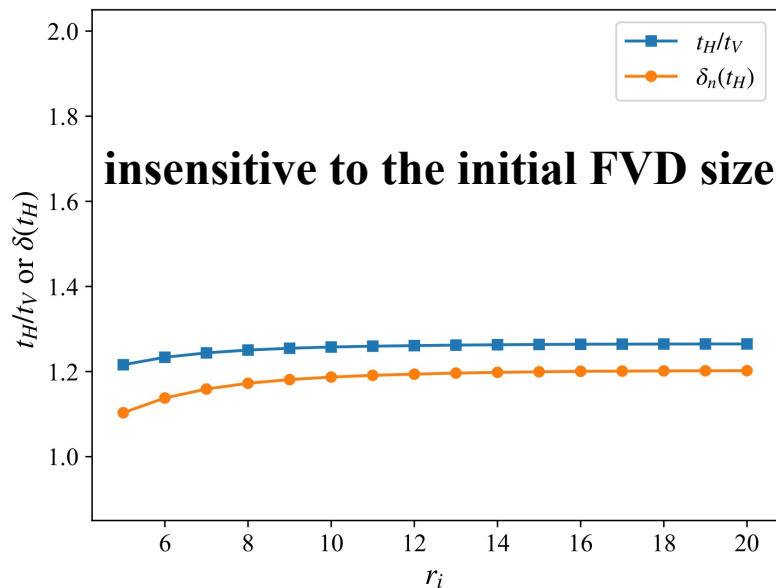
- Horizon crossing time $t_H = \frac{r_i^2}{4t_i}$
- Vacuum energy domination $t_V = \frac{1}{2} \sqrt{\frac{3}{8\pi\rho_V}}$
- $t_H/t_V \gtrsim 1$: type B PBH
- $t_H/t_V \lesssim 1$: type A PBH or dispersion

2. Local density contrast $\delta(t_H)$

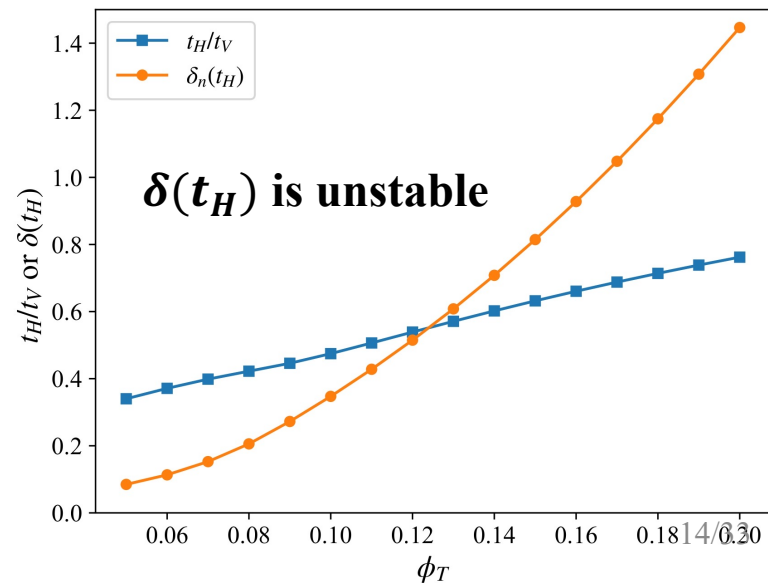
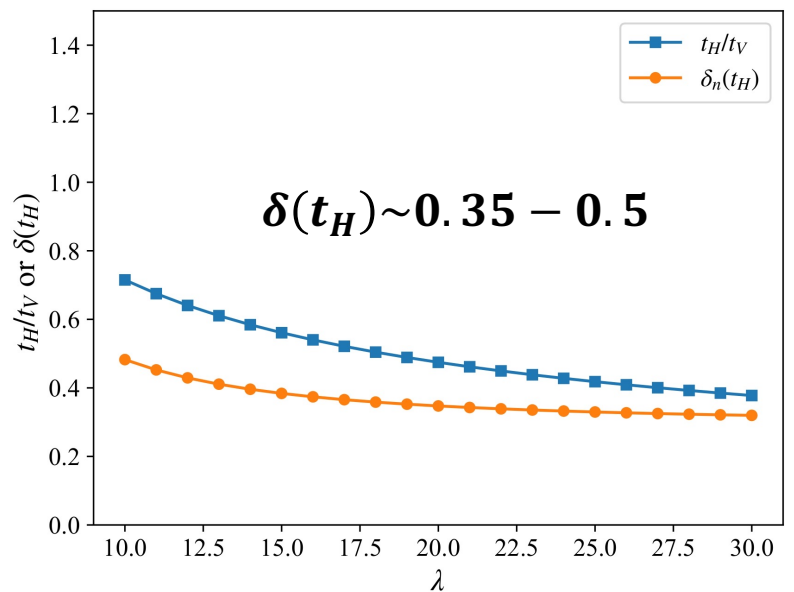
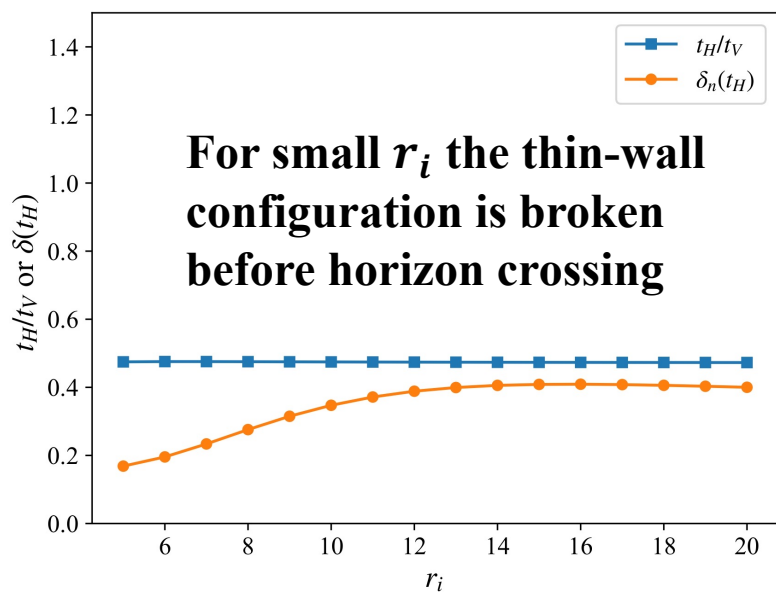
$$\delta(t_H) = \frac{\rho_{\text{in}}(t_H) - \rho_{\text{out}}(t_H)}{\rho_{\text{out}}(t_H)}$$



Threshold for type B PBH



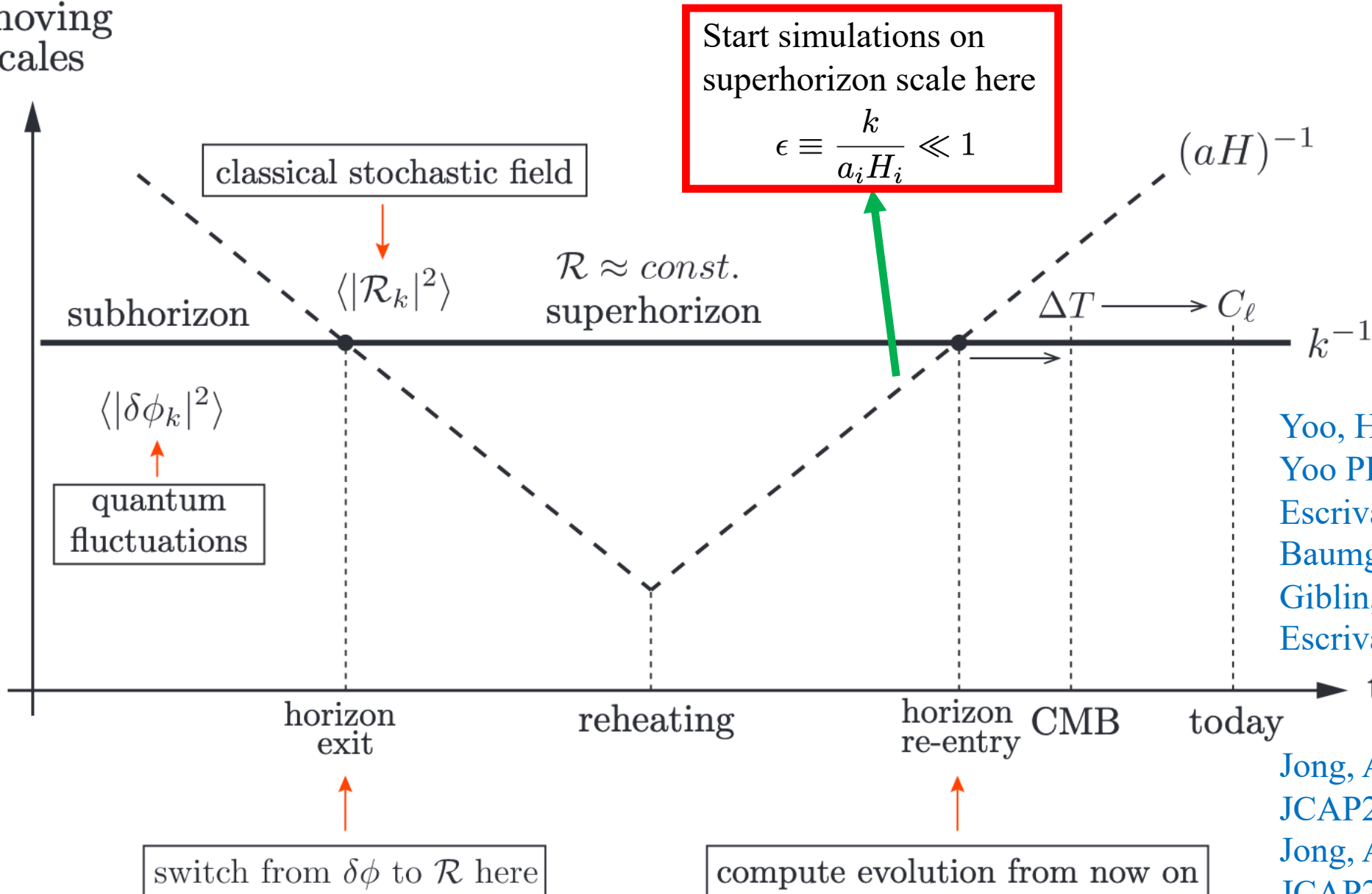
Threshold for type A PBH



PBH Formation from Curvature Perturbations in 3D

PBH formation from curvature perturbations

comoving
scales



Most papers have imposed spherical symmetry, and **fully 3+1 simulations** remain much less common

Yoo, Harada, Okawa PRD2020;
Yoo PRD 2024;
Escrivà, Yoo PRD 2025;
Baumgarte, Clough, Gerhardinger,
Giblin, Miller 2606.30641
Escrivà 2609.03051

Jong, Aurrekoetxea, Lim
JCAP2022
Jong, Aurrekoetxea, Lim, Franca
JCAP2023

Why is this a hard 3D numerical problem?

➤ Spatial hierarchy

- Start with a perturbation well outside the Hubble radius
- After re-entry, resolve a much smaller collapsing core
- Near criticality, the newly formed black hole becomes even smaller

➤ Temporal hierarchy

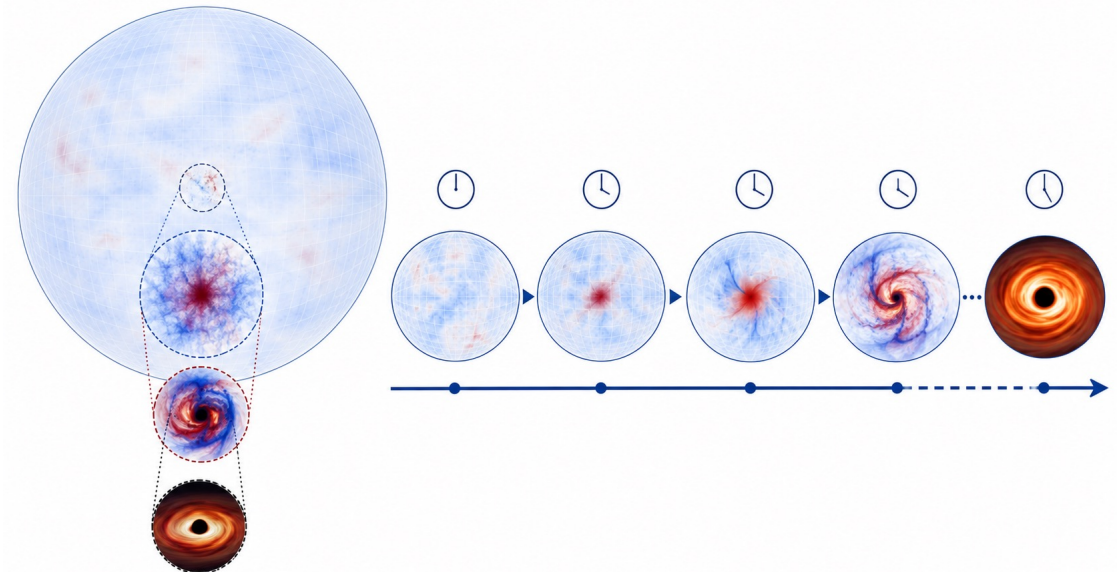
- We need time for horizon re-entry
- Formation is not the end: we want many-Hubble-time post-formation evolution

[Yoo et al. JOSS2026]

COSMOS: A 3D numerical relativity code specialized for PBH formation

- scale-up coordinates and fixed mesh refinement

Adaptive mesh refinement (AMR) provides another natural way



Formulation

Radiation-dominated Universe: Einstein gravity + radiation fluid

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

BSSN formalism

Built-in in GRChombo

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt) (dx^j + \beta^j dt)$$

$$\chi \equiv \gamma^{-1/3}, \quad \tilde{\gamma}_{ij} \equiv \chi \gamma_{ij},$$

$$K \equiv \gamma^{ij} K_{ij}, \quad \tilde{A}_{ij} \equiv \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right),$$

$$\tilde{\Gamma}^i \equiv \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i, \quad \gamma \equiv \det(\gamma_{ij})$$

$$E \equiv n_\mu n_\nu T^{\mu\nu}, \quad \text{Eulerian (ADM) energy density}$$

$$S_i \equiv -\gamma_{i\mu} n_\nu T^{\mu\nu},$$

$$S_{ij} \equiv \gamma_{i\mu} \gamma_{j\nu} T^{\mu\nu},$$

$$S \equiv \gamma^{ij} S_{ij}.$$

Fourth-order finite difference method

$$\partial_t \mathcal{E} + \partial_j [\mathcal{E}(\alpha v^j - \beta^j) + \alpha \sqrt{\gamma} p v^j] = \mathcal{Q}_E,$$

$$\partial_t \mathcal{S}_i + \partial_j [\mathcal{S}_i(\alpha v^j - \beta^j) + \alpha \sqrt{\gamma} p \delta_i^j] = \mathcal{Q}_{S_i},$$

where

$$\mathcal{Q}_E = \alpha \sqrt{\gamma} S^{ij} K_{ij} - \sqrt{\gamma} S^i \partial_i \alpha,$$

$$\mathcal{Q}_{S_i} = -\mathcal{E} \partial_i \alpha + \mathcal{S}_j \partial_i \beta^j + \frac{1}{2} \alpha \sqrt{\gamma} S^{jk} \partial_i \gamma_{jk}.$$

$$\nabla_\mu T^{\mu\nu} = 0$$

Relativistic hydrodynamics

Our implementation

$$T^{\mu\nu} = (\rho + p) u^\mu u^\nu + p g^{\mu\nu}$$

ρ : proper energy density

$$u^\mu = W(n^\mu + v^\mu), \quad \text{with} \quad n_\mu v^\mu = 0,$$

$$W = (1 - v^2)^{-1/2},$$

$$E = (\rho + p) W^2 - p,$$

$$S_i = (\rho + p) W^2 v_i,$$

$$S_{ij} = (\rho + p) W^2 v_i v_j + p \gamma_{ij}.$$

Flux-conservative discretization

Flux-conservative relativistic hydrodynamics

Equation of state: $p = \omega \rho, \omega = 1/3$

- No separate baryon rest-mass density is needed
- Primitive variables: (ρ, v^i)
- Conserved variables: densitized energy and momentum $\mathcal{E} \equiv \sqrt{\gamma} E, \quad \mathcal{S}_i \equiv \sqrt{\gamma} S_i$
- ✓ Evolve densitized energy and momentum only
- ✓ Conserved-to-primitive recovery is analytical
- ✓ MUSCL (monotonic upstream-centered scheme for conservation laws) reconstruction of primitive variables [Use cell-center data to reconstruct cell-face data]
- ✓ Local Lax–Friedrichs numerical flux at cell faces
- ✓ Fourth-order Runge–Kutta for the coupled BSSN + fluid system

Superhorizon initial data

$$\gamma_{ij} = a_i^2 e^{2\zeta(\mathbf{x})} \delta_{ij} + \mathcal{O}(\epsilon^2), \quad \epsilon \equiv \frac{k}{a_i H_i} \ll 1,$$

$$\zeta(r) = \mu \exp(-k^2 r^2) W(r)$$

Long-wavelength /
gradient-expansion
growing mode

[Harada et al. PRD2015]

$$\chi = a_i^{-2} e^{-2\zeta} \left[1 + \frac{2f}{3(1+\omega)} \frac{1}{(a_i H_i)^2} \right] + \mathcal{O}(\epsilon^4), \quad (17a)$$

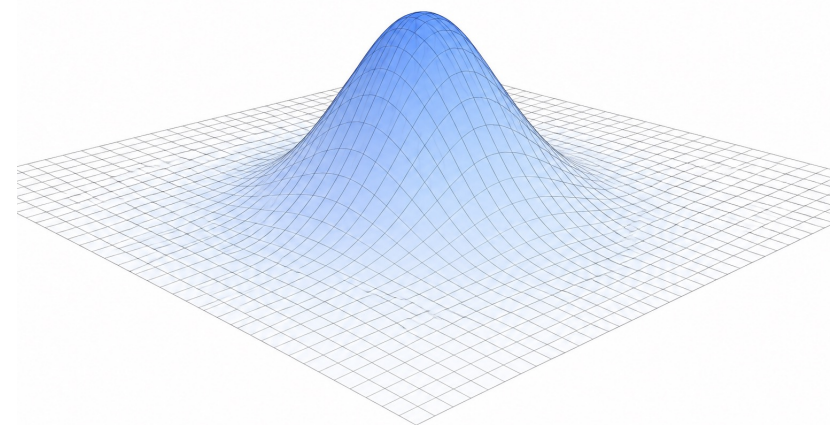
$$\tilde{\gamma}_{ij} = \delta_{ij} - \frac{4p_{ij}}{(1+3\omega)(5+3\omega)} \frac{1}{(a_i H_i)^2} + \mathcal{O}(\epsilon^4), \quad (17b)$$

$$K = -3H_i, \quad (17c)$$

$$\tilde{A}_{ij} = \frac{2H_i p_{ij}}{5+3\omega} \frac{1}{(a_i H_i)^2} + \mathcal{O}(\epsilon^4), \quad (17d)$$

$$\alpha = 1 - \frac{1+3\omega}{3(1+\omega)} f \frac{1}{(a_i H_i)^2} + \mathcal{O}(\epsilon^4), \quad \beta^i = 0. \quad (17e)$$

- Fluid variables are fixed algebraically using the Hamiltonian and momentum constraints
- Characteristic scale $r_m = \frac{1}{k}$
- Defining horizon entry by $a(t_H)r_m = H^{-1}(t_H)$ and horizon mass $M_H = 1/2H(t_H)$

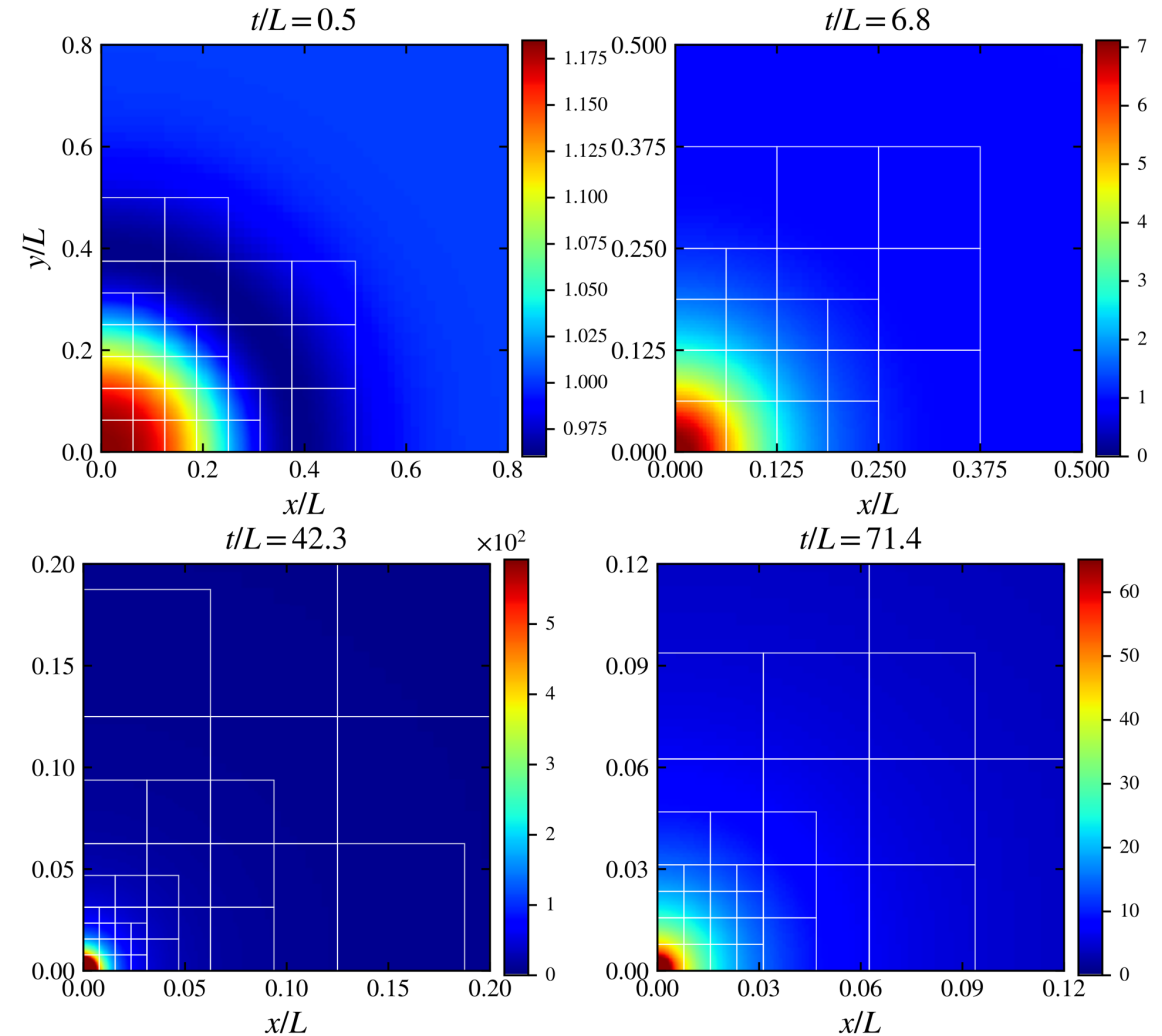
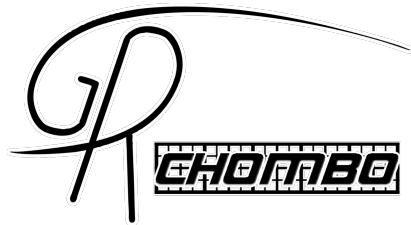


μ sets the perturbation amplitude and is the control parameter for collapse

AMR solves the spatial hierarchy

- GRChombo / Chombo block-structured adaptive mesh refinement
- Reflection conditions at both boundaries of a octant: equivalent to a periodic box here
- Refine on gradients of conformal factor χ and energy density E

AMR addresses the **spatial** bottleneck.
The remaining problem is **temporal**



Cosmic-time slicing is physically natural

$$(\partial_t - \beta^i \partial_i) \alpha = -2\alpha(K - K_b)$$

- $K_b = -3H(t)$ subtracts the homogeneous FLRW expansion
- Far from the perturbation: $\alpha \rightarrow 1$, so coordinate time $\approx$ cosmic time
- Near collapse: local $K - K_b$ drives lapse collapse, retaining singularity avoidance

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j$$

$$\frac{dx}{dt} = \frac{v}{a(t)}$$

Light, sound and lapse-gauge modes propagate ever more slowly in comoving coordinates

$$\text{physical CFL} \Rightarrow \Delta t \propto a(t) \Delta x$$

In principle, the allowed cosmic-time step should grow with the scale factor

The standard Gamma-driver obstructs $\Delta t \propto a\Delta x$

$$\partial_t \beta^i = bB^i, \quad \partial_t B^i = \partial_t \tilde{\Gamma}^i - \eta_{\text{GD}} B^i$$

On an FLRW background, the shift-dependent principal part is built from the conformal metric,
not the physical metric

$$v_T = \sqrt{b}, \quad v_L = \sqrt{\frac{4b}{3}}$$

standard choice $b = \frac{3}{4} \Rightarrow v_L = 1$

physical modes: $v_{\text{coord}} \propto a^{-1}$

shift gauge: $v_{\text{coord}} = \text{const}$

The shift CFL therefore imposes a fixed global time-step cap even when the physical CFL becomes less restrictive

A separate explicit–integration bound also comes from the damping term: $\eta_{\text{GD}} \Delta t \lesssim \text{const}$

A cosmologically scaled Gamma-driver

Choose the driver coefficients so that its characteristic and damping time scales redshift with the background

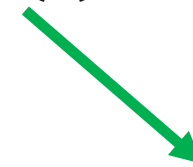
$$b(t) = b_0 \left(\frac{a_i}{a(t)} \right)^2$$

$$v_{\text{shift}} \propto \sqrt{b} \propto a^{-1}$$

$$\eta_{\text{GD}}(t) = \eta_{\text{GD},0} \left(\frac{a_i}{a(t)} \right)$$

$$\eta_{\text{GD}} \Delta t = \text{const} \quad \text{when} \quad \Delta t \propto a$$

$$\Delta t_0 = \min \left[C_{\text{CFL}} a(t) \Delta x_0, \frac{C_H}{H(t)} \right]$$



limits fractional background expansion per coarse-level step

Key point: keep cosmic time, but take an approximately constant conformal-time step

Independent check: evolve directly in conformal time

$$dt = a d\eta$$

$$\alpha_\eta = a\alpha_t, \quad \beta_\eta^i = a\beta_t^i$$

FLRW metric in conformal time

$$ds^2 = a^2(t)(-dt^2 + d\vec{x}^2)$$

Conformal Hubble parameter

$$\mathcal{H} = a'/a$$

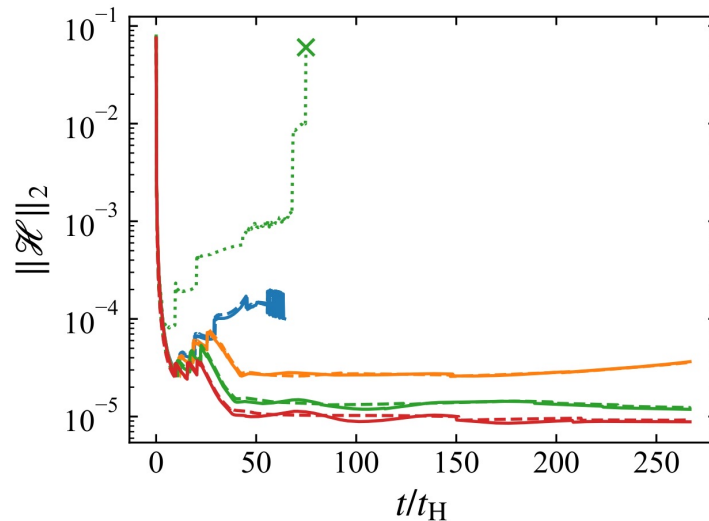
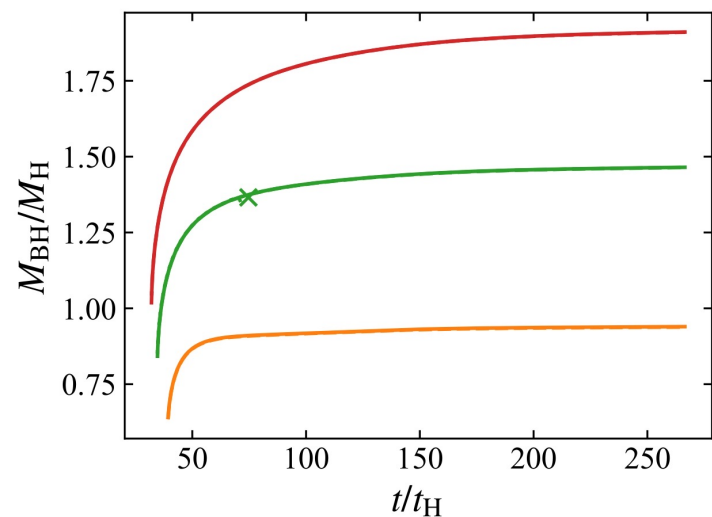
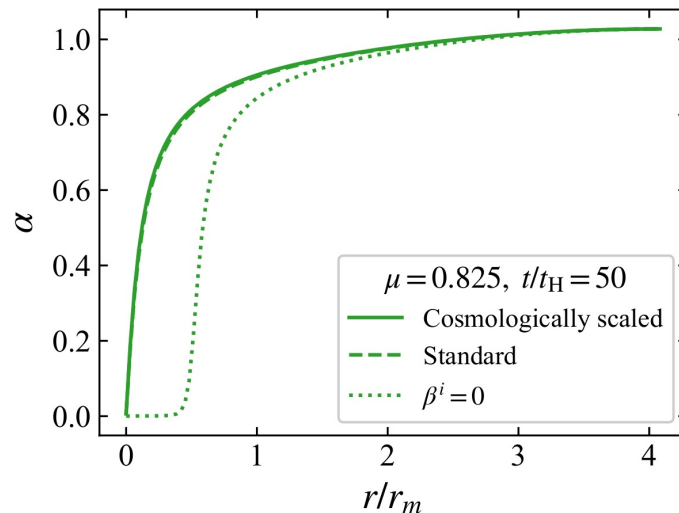
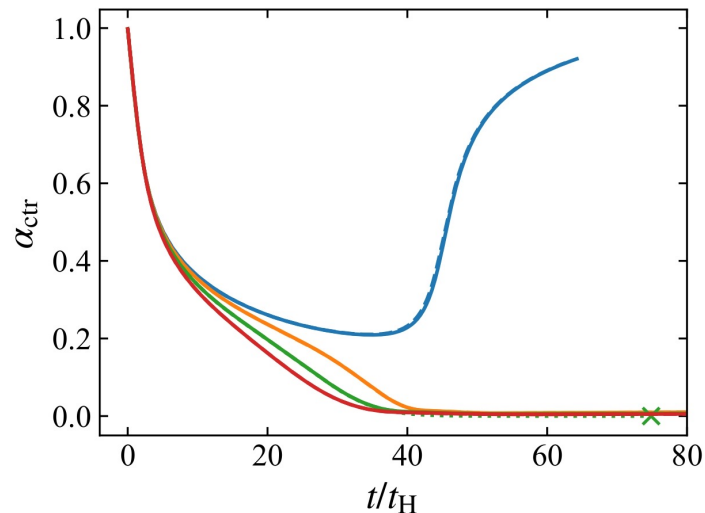
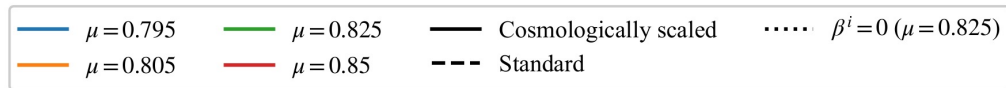
conformal–time gauge

$$(\partial_\eta - \beta_\eta^i \partial_i) \alpha_\eta = \mathcal{H} \alpha_\eta - 2a \alpha_\eta (K - K_b)$$

The main results below use the scaled cosmic-time construction for direct comparison with previous PBH simulations.

- In conformal time, light and sound have approximately constant coordinate speeds in FLRW
- The standard constant-coefficient Gamma-driver can then be used
- This is an independent gauge / implementation check — not an exact reparametrization of the scaled driver

Gauge performance and strong-field robustness



- Scaled and standard drivers give nearly identical collapse histories and horizon masses
- Constraint histories remain of the same magnitude through black-hole formation
- A frozen shift agrees initially but eventually fails from coordinate stretching

The scaled driver keeps the strong-field moving-puncture behaviour while removing the cosmological time-step bottleneck

The payoff: about 94× less computational cost

Representative run: $\mu = 0.825$, evolved to $\frac{t}{t_H} \simeq 266.8$

102,436

standard driver



1,092

scaled driver

$\approx 94\times$ reduction in coarse–level advances

$\approx 94\times$ reduction in computational cost

The apparent-horizon mass evolution and constraint behaviour remain consistent

Collapse threshold: a precise 3D benchmark

At $\Delta x_{\min} = \frac{L}{4096}$, an amplitude scan gives

$$0.79578 < \mu_c < 0.79580$$

For the same Gaussian curvature profile:

3D COSMOS: $0.795 < \mu_c < 0.805$

Spherical Misner–Sharp: $\mu_c \simeq 0.79579 \pm 10^{-5}$

Why this matters

A stringent test of the 3D fluid evolution

Confirms that Cartesian BSSN + GRHD + AMR reaches the correct collapse boundary

Black-hole formation is defined by a marginally outer trapped surface, not by the gauge-dependent lapse alone

Critical mass scaling in 3D

$$M_{\text{BH}} = M_H K (\mu - \mu_c)^\gamma$$

Joint fit to the 10 nearest supercritical points:

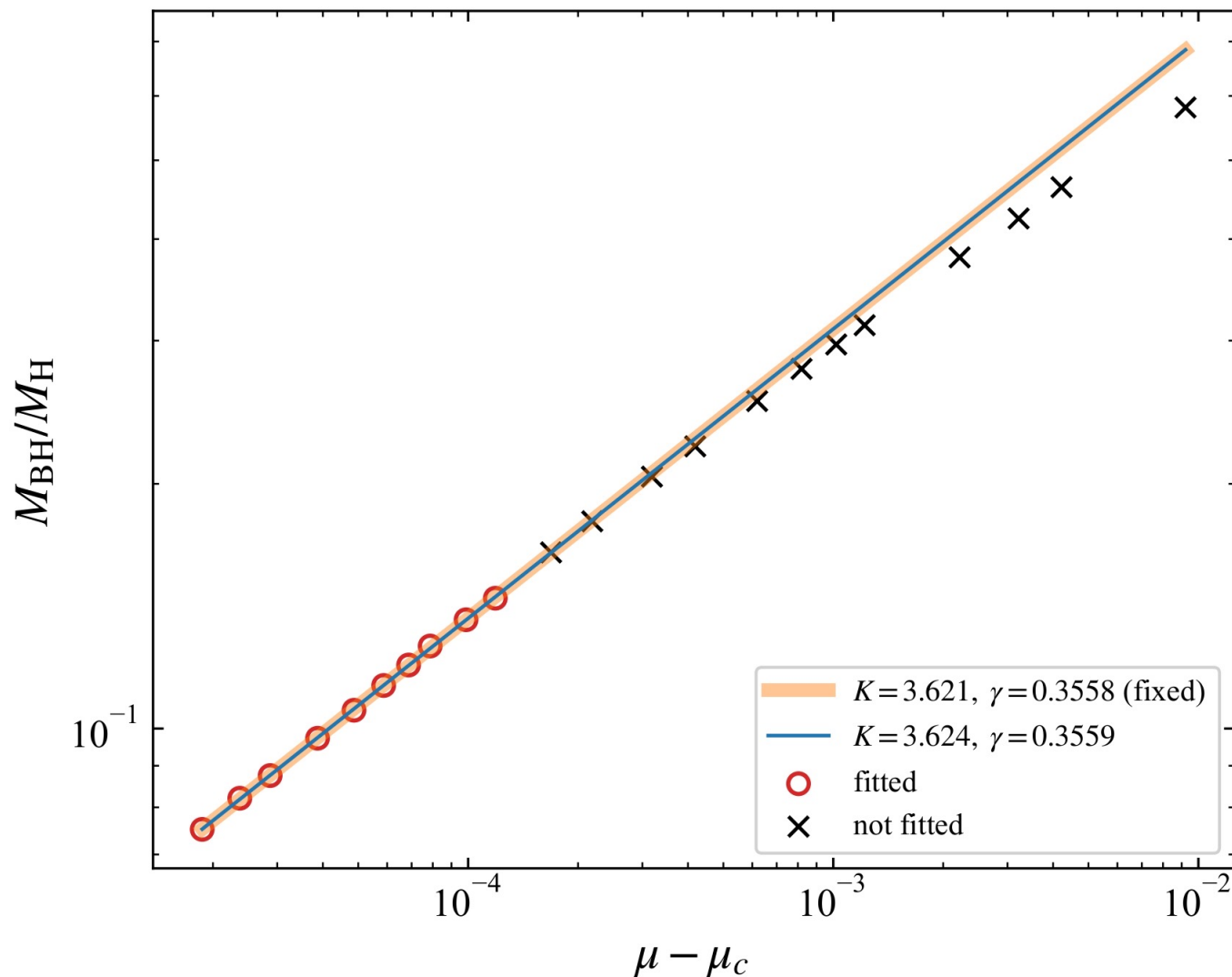
$$K \simeq 3.624$$

$$\mu_c \simeq 0.7957813$$

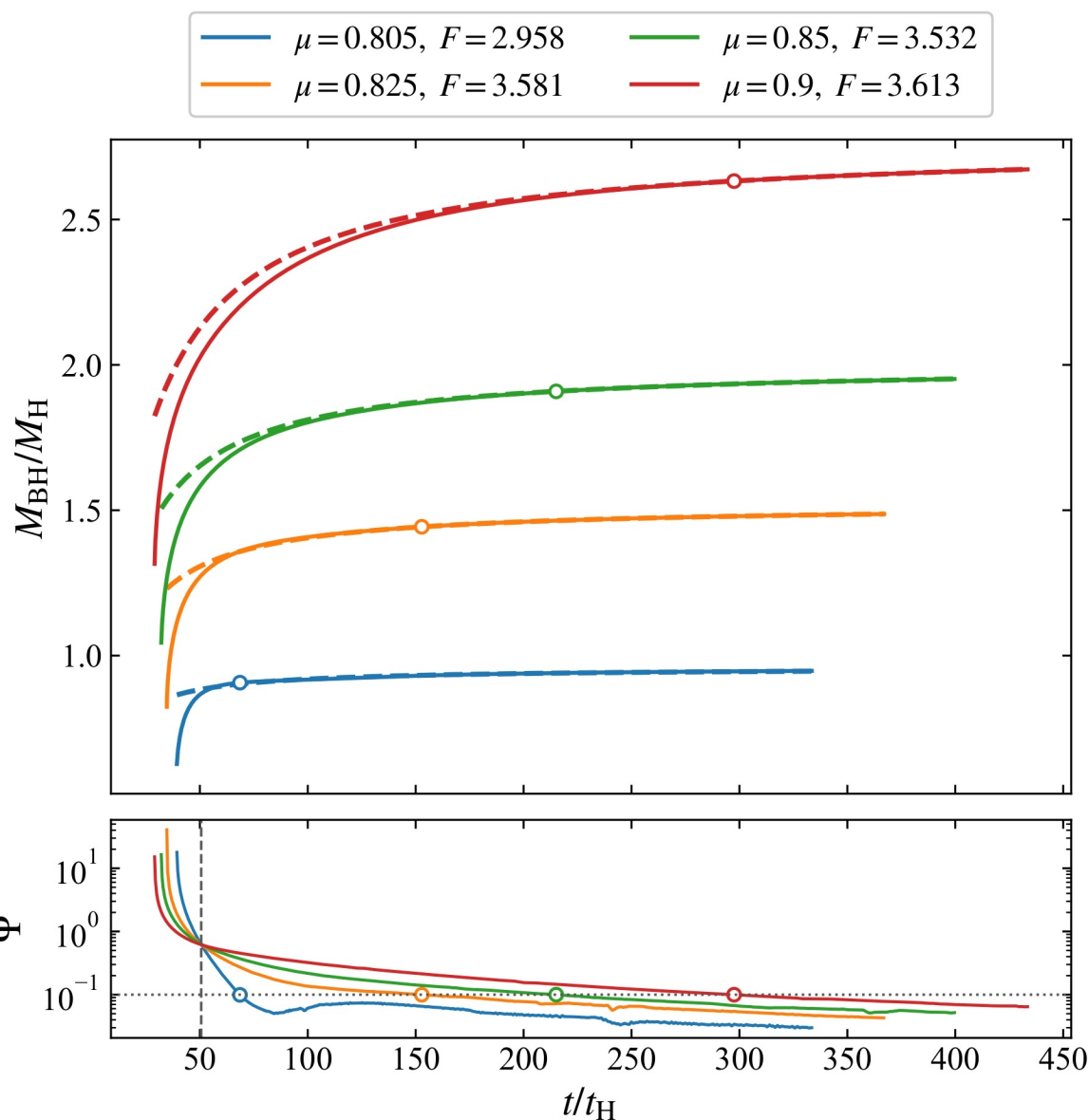
$$\gamma \simeq 0.3559$$

Known radiation-fluid exponent: $\gamma \simeq 0.3558$

**A nontrivial strong-field benchmark: the 3D
Cartesian code recovers the universal
spherical critical behaviour**



Long after formation: PBH mass growth



$$\dot{M}_{BH} = \frac{3F}{2} \frac{M_{BH}^2}{t^2}$$

$$M_{BH}(t) = \left[\frac{1}{M_\infty} + \frac{3F}{2t} \right]^{-1}$$

Fit only once $\Psi \equiv \frac{\dot{M}_{BH}}{HM_{BH}} \leq 0.1$, i.e. in the late-time regime

TABLE I. Late-time accretion fits using the contiguous data from the first point with $\Psi \leq 0.1$ to the end of each run.

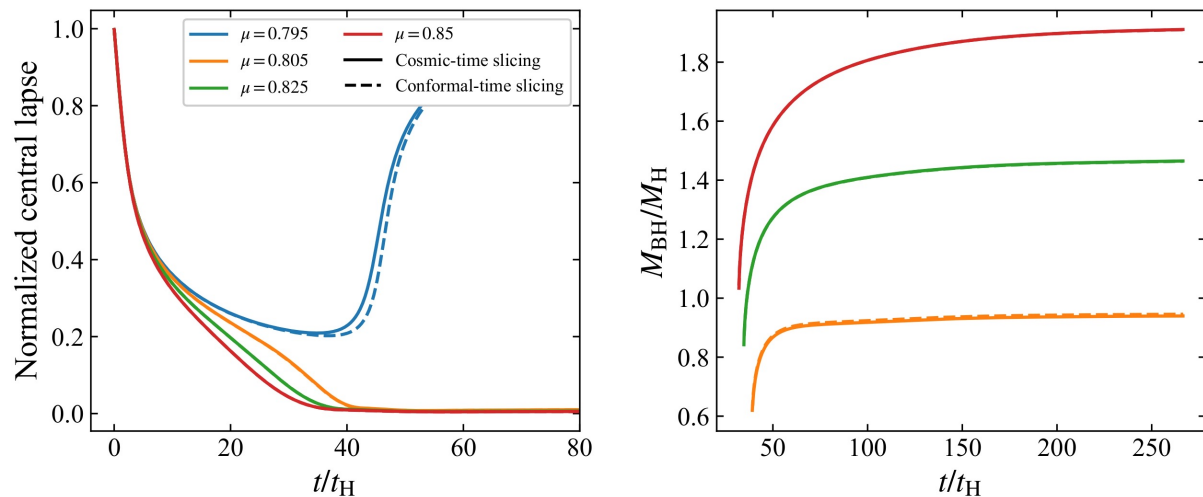
μ	F	M_∞/M_H
0.805	2.958	0.9584
0.825	3.581	1.5201
0.850	3.532	2.0035
0.900	3.613	2.7651

Broadly consistent with previous spherical Misner–Sharp results

This is why the long-time gauge acceleration is useful physically, not just computationally

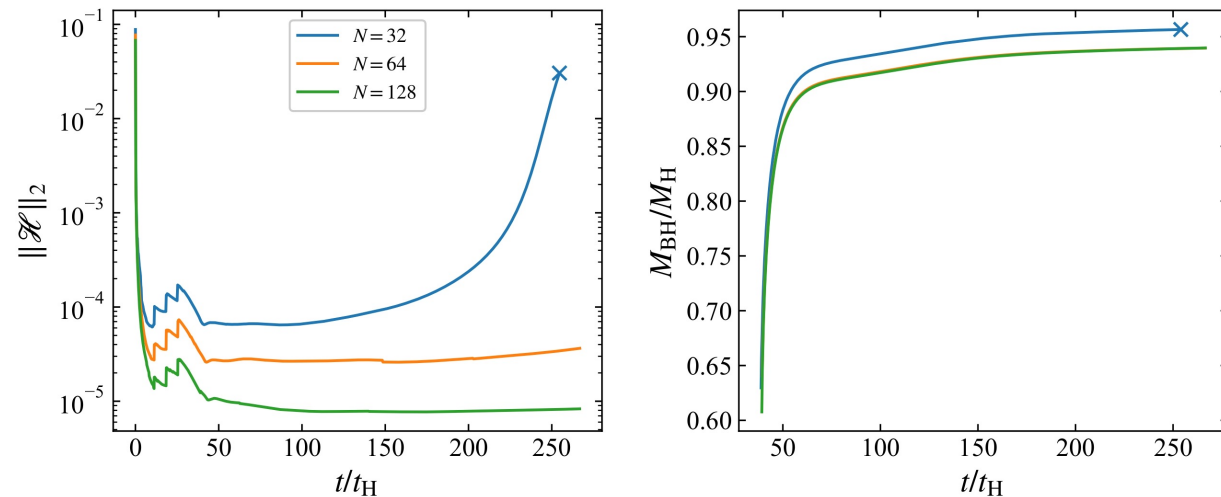
Independent checks: gauge choice and resolution

Conformal-time vs scaled cosmic-time gauge



- Horizon masses agree between the two independent gauge constructions
- Representative advance counts: 1092 (cosmic-time) vs 1090 (conformal-time)

Resolution study



- Hamiltonian-constraint norm decreases systematically with resolution
- Medium- and high-resolution horizon masses closely agree

Taken together: gauge and resolution checks all support the exterior observables used in the physics analysis

Conclusions

- **PBH formation from FVDs**: Identified three outcomes (type B vs type A vs dispersal)
- PBH formation criteria: Validated t_H/t_V as a more robust criterion than $\delta(t_H)$

Threshold	t_H/t_V	$\delta(t_H)$
Type B	$\sim 1.1 - 1.6$	$\sim 1.0 - 1.7$ (unstable)
Type A	$\sim 0.35 - 0.7$	$\sim 0.35 - 0.5$ (unstable)

- **A new 3D simulation framework for PBH formation** based on AMR code GRChombo
- **Cosmologically scaled Gamma-driver** removes the cosmological gauge-CFL bottleneck
- Validation against a spherical Gaussian curvature profile: threshold, critical scaling, and late-time accretion
- Next: nonspherical perturbations, PBH spin, multi-peak collapse, and environmental effects

Thank you!

Backup: BSSN variables and matter source terms

$$\chi = \gamma^{-\frac{1}{3}}, \quad \tilde{\gamma}_{ij} = \chi \gamma_{ij}, \quad K = \gamma^{ij} K_{ij}, \quad \tilde{A}_{ij} = \chi (K_{ij} - \frac{1}{3} \gamma_{ij} K)$$

$$E = n_\mu n_\nu T^{\mu\nu}, \quad S_i = -\gamma_{i\mu} n_\nu T^{\mu\nu}, \quad S_{ij} = \gamma_{i\mu} \gamma_{j\nu} T^{\mu\nu}$$

$$\mathcal{H} = R^{(3)} + K^2 - K_{ij} K^{ij} - 16\pi E = 0$$

$$\mathcal{M}_i = D_j K_i^j - D_i K - 8\pi S_i = 0$$

The spacetime sector is the standard GRChombo BSSN infrastructure; this work supplies the matter source terms and conservative fluid evolution

Backup: analytic conserved-to-primitive recovery

$$\mathcal{E} = \sqrt{\gamma} E, \quad \mathcal{S}_i = \sqrt{\gamma} S_i \quad S^2 = \gamma^{ij} S_i S_j$$

$$\omega \rho^2 + (1 - \omega) E \rho - (E^2 - S^2) = 0$$

$$\rho = \frac{-(1 - \omega)E + \sqrt{(1 - \omega)^2 E^2 + 4\omega(E^2 - S^2)}}{2\omega}$$

$$p = \omega \rho, \quad v^i = \frac{S^i}{E + p}, \quad W = (1 - v^2)^{-\frac{1}{2}}$$

For the radiation EOS, no iterative root finder is required

Backup: MUSCL reconstruction

primitive variable $q \in \{\rho, v^1, v^2, v^3\}$ at grid point I

$$(\Delta_+ q)_I = q_{I+1} - q_I, \quad (\Delta_- q)_I = q_I - q_{I-1}, \quad (\text{A5})$$

and the limited differences by

$$\begin{aligned} (\bar{\Delta}_+ q)_I &= \text{minmod}[(\Delta_+ q)_I, b_{\text{lim}}(\Delta_- q)_I], \\ (\bar{\Delta}_- q)_I &= \text{minmod}[(\Delta_- q)_I, b_{\text{lim}}(\Delta_+ q)_I]. \end{aligned} \quad (\text{A6})$$

Here, the minmod function is defined as

$$\text{minmod}(x, y) = \begin{cases} \text{sgn}(x) \min(|x|, |y|), & xy > 0, \\ 0, & xy \leq 0. \end{cases} \quad (\text{A7})$$

The left and right states at the face $I + 1/2$ are reconstructed according to

$$\begin{aligned} q_{I+1/2}^L &= q_I + \frac{1-\kappa}{4}(\bar{\Delta}_- q)_I + \frac{1+\kappa}{4}(\bar{\Delta}_+ q)_I, \\ q_{I+1/2}^R &= q_{I+1} - \frac{1-\kappa}{4}(\bar{\Delta}_+ q)_{I+1} - \frac{1+\kappa}{4}(\bar{\Delta}_- q)_{I+1}. \end{aligned} \quad (\text{A8})$$

We set $\kappa = 1/3$ and choose the limiter parameter $b_{\text{lim}} = 2$.

The geometric and gauge variables are interpolated to the same face using fourth-order midpoint interpolation.

We then convert the reconstructed primitive states to the densitized conserved states $\mathbf{U}_{L,R} = (\mathcal{E}, \mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3)_{L,R}^T$ using the common face-centered geometry.

Backup: local Lax-Friedrichs flux

For a face normal to the x^k direction, the components of the physical flux $\mathbf{F}^{(k)} = (F_{\mathcal{E}}^{(k)}, F_{\mathcal{S}_1}^{(k)}, F_{\mathcal{S}_2}^{(k)}, F_{\mathcal{S}_3}^{(k)})^\top$ are

$$\begin{aligned} F_{\mathcal{E}}^{(k)} &= \mathcal{E}(\alpha v^k - \beta^k) + \alpha \sqrt{\gamma} p v^k, \\ F_{\mathcal{S}_m}^{(k)} &= \mathcal{S}_m(\alpha v^k - \beta^k) + \alpha \sqrt{\gamma} p \delta_m^k, \quad m = 1, 2, 3. \end{aligned} \quad (\text{A9})$$

We combine the left and right states using the local Lax–Friedrichs flux,

$$\hat{\mathbf{F}}^{(k)} = \frac{1}{2} \left[\mathbf{F}^{(k)}(\mathbf{U}_L) + \mathbf{F}^{(k)}(\mathbf{U}_R) - a_\star^{(k)}(\mathbf{U}_R - \mathbf{U}_L) \right]. \quad (\text{A10})$$

The coordinate characteristic speeds are

$$\begin{aligned} \lambda_0^{(k)} &= \alpha v^k - \beta^k, \\ \lambda_\pm^{(k)} &= -\beta^k + \frac{\alpha}{1 - \omega v^2} \left[(1 - \omega) v^k \pm \sqrt{\mathcal{D}_k} \right], \end{aligned} \quad (\text{A11})$$

where $\mathcal{D}_k \equiv \omega(1 - v^2) [\gamma^{kk}(1 - \omega v^2) - (1 - \omega)(v^k)^2]$. We evaluate the local propagation speed from both reconstructed states as

$$a_\star^{(k)} = \max_{A \in \{L, R\}} \left\{ \left| \lambda_{0,A}^{(k)} \right|, \left| \lambda_{+,A}^{(k)} \right|, \left| \lambda_{-,A}^{(k)} \right| \right\}. \quad (\text{A12})$$

Only the extremal eigenvalues are required; the method does not rely on a complete characteristic decomposition or an exact Riemann solver.

The semi-discrete flux-conservative evolution equation for a cell-centered conserved state is

$$\frac{d\mathbf{U}_I}{dt} = - \sum_{k=1}^3 \frac{\hat{\mathbf{F}}_{I+\mathbf{e}_k/2}^{(k)} - \hat{\mathbf{F}}_{I-\mathbf{e}_k/2}^{(k)}}{\Delta x_k} + \mathbf{Q}_I, \quad (\text{A13})$$

where $\mathbf{e}_k$ is the unit grid vector in the x^k direction and $\mathbf{Q} = (\mathcal{Q}_E, \mathcal{Q}_{S_i})$ collects the cell-centered geometric source terms. We recompute the primitive variables, reconstructed states, fluxes, and source terms at every Runge–Kutta substep. At coarse–fine AMR interfaces, each refinement level evaluates its fluxes independently.

Backup: shift-mode characteristic scaling

$$\partial_t \tilde{\Gamma}^i \simeq \Delta \beta^i + \frac{1}{3} \partial^i \partial_j \beta^j$$

$$\partial_t^2 \beta^i + \eta_{\text{GD}} \partial_t \beta^i \simeq b [\Delta \beta^i + \frac{1}{3} \partial^i \partial_j \beta^j]$$

$$\omega^2 + i\eta_{\text{GD}}\omega = bk^2 \quad (\text{transverse})$$

$$\omega^2 + i\eta_{\text{GD}}\omega = \frac{4b}{3}k^2 \quad (\text{longitudinal})$$

$$v_T = \sqrt{b}, \quad v_L = \sqrt{\frac{4b}{3}}$$

$$b \propto a^{-2} \Rightarrow v_{\text{shift}} \propto a^{-1}$$

Time dependence adds only a lower-order expansion-damping term; it does not change the principal characteristic speeds

Backup: time-step prescription

$$\Delta t_0 = \min[C_{\text{CFL}} a(t) \Delta x_0, \frac{C_H}{H(t)}]$$

$$C_{\text{CFL}} = 0.25, \quad C_H = 0.02$$

- First term: physical + gauge CFL condition with redshifting coordinate speeds
- Second term: limits fractional background expansion per coarse-level step
- Berger–Olinger subcycling: $\Delta t_\ell = \frac{\Delta t_0}{2^\ell}$, so $\frac{\Delta t_\ell}{\Delta x_\ell}$ is constant across levels

Backup: interior regularization prescription

$$\text{RHS} \rightarrow w_{\text{reg}}(r) \times \text{RHS} \quad w_{\text{reg}} = \exp\left(-s\left[1 - \frac{r^2}{r_{\text{reg}}^2}\right]^p\right) \quad (r < r_{\text{reg}})$$

$$r_{\text{reg}} = f_{\text{reg}} r_{\text{AH,min}}, \quad f_{\text{reg}} = 0.5, \quad p = 4$$

- Applied only after the first apparent horizon is found
- Does not remove cells or impose an inner boundary
- Outside r_{reg} , the evolution equations and numerical update are unchanged
- Tested with $s = 4, 6, 8$; exterior fields and horizon masses remain insensitive

Backup: fiducial numerical parameters

Initial Hubble rate: $H_i = 50$

Gaussian scale: $k = \frac{10}{\sqrt{6}}$

Domain half-size: $L = 1$

Window radius: $r_W = 0.8L$

Characteristic radius: $r_m = \frac{\sqrt{6}}{10}$

$t_i = 0.01, \quad t_H = M_H = 1.5$

Octant base grid: 64^3

5 additional AMR levels

Finest $\Delta x = \frac{L}{2048}$

Long accretion runs: $L = 2$ at fixed physical resolution

Apparent-horizon mass: $M_{BH} = \sqrt{\frac{A_{AH}}{16\pi}}$

Backup: key numbers at a glance

collapse threshold

$$0.79578 < \mu_c < 0.79580$$

critical exponent

$$\gamma \simeq 0.3559$$

scaled-driver speedup

$$\approx 94\times \text{fewer coarse advances}$$

late-time efficiencies

$$F \simeq 3.53\text{--}3.61 \quad (\text{larger } \mu)$$

All numbers above are for the spherical Gaussian profile used as the first validation application of the 3D framework